

Answer ALL questions. Write your answers in the spaces provided.

1. A chocolate manufacturer places special tokens in 2% of the bars it produces so that each bar contains at most one token. Anyone who collects 3 of these tokens can claim a prize.

Andreia buys a box of 40 bars of the chocolate.

- (a) Find the probability that Andreia can claim a prize.

(2)

Barney intends to buy bars of the chocolate, one at a time, until he can claim a prize.

- (b) Find the probability that Barney can claim a prize when he buys his 40th bar of chocolate.

(3)

- (c) Find the expected number of bars that Barney must buy to claim a prize.

(1)

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2. Indre works on reception in an office and deals with all the telephone calls that arrive. Calls arrive randomly and, in a 4-hour morning shift, there are on average 80 calls.

- (a) Using a suitable model, find the probability of more than 4 calls arriving in a particular 20-minute period one morning.

(3)

Indre is allowed 20 minutes of break time during each 4-hour morning shift, which she can take in 5-minute periods. When she takes a break, a machine records details of any call in the office that Indre has missed.

One morning Indre took her break time in 4 periods of 5 minutes each.

- (b) Find the probability that in exactly 3 of these periods there were no calls.

(2)

On another occasion Indre took 1 break of 5 minutes and 1 break of 15 minutes.

- (c) Find the probability that Indre missed exactly 1 call in each of these 2 breaks.

(3)

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1. The number of customers entering Jeff's supermarket each morning follows a Poisson distribution.

Past information shows that customers enter at an average rate of 2 every 5 minutes.

Using this information,

- (a) (i) find the probability that exactly 26 customers enter Jeff's supermarket during a randomly selected 1-hour period one morning, (2)
- (ii) find the probability that at least 21 customers enter Jeff's supermarket during a randomly selected 1-hour period one morning. (2)

A rival supermarket is opened nearby. Following its opening, the number of customers entering Jeff's supermarket over a randomly selected 40-minute period is found to be 10

- (b) Test, at the 5% significance level, whether or not there is evidence of a decrease in the rate of customers entering Jeff's supermarket. State your hypotheses clearly. (4)

A further randomly selected 20-minute period is observed and the hypothesis test is repeated. Given that the true rate of customers entering Jeff's supermarket is now 1 every 5 minutes,

- (c) calculate the probability of a Type II error. (5)



2. The discrete random variables W , X and Y are distributed as follows

$$W \sim B(10, 0.4)$$

$$X \sim \text{Po}(4)$$

$$Y \sim \text{Po}(3)$$

(a) Explain whether or not $\text{Po}(4)$ would be a good approximation to $B(10, 0.4)$

(1)

(b) State the assumption required for $X + Y$ to be distributed as $\text{Po}(7)$

(1)

Given the assumption in part (b) holds,

(c) find $P(X + Y < \text{Var}(W))$

(2)

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- (f) Carry out the test using a 5% level of significance. (4)

3. A courier delivers parcels. The random variable X represents the number of parcels delivered successfully each day by the courier where $X \sim B(400, 0.64)$

A random sample $X_1, X_2, \dots, X_{100}$ is taken.

Estimate the probability that the mean number of parcels delivered each day by the courier is greater than 257

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- (2)

She records the number of calls for help received on each of these days.

- (4)

(4)

(3)

- (1)

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins or other markings on the paper.

2. The number of errors made by a secretary is modelled by a Poisson distribution with a mean of 2.4 per 100 words.

A 100-word piece of work completed by the secretary is selected at random.

- (a) Find the probability that

(i) there are exactly 3 errors,

(ii) there are fewer than 2 errors.

(2)

After a long holiday, a randomly selected piece of work containing 250 words completed by the secretary is examined to see if the rate of errors has changed.

- (b) Stating your hypotheses clearly, and using a 5% level of significance, find the critical region for a suitable test.

(4)

- (c) Find $P(\text{Type I error})$ for the test in part (b)

(1)

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7. The probability of winning a prize when playing a single game of *Pento* is $\frac{1}{5}$

When more than one game is played the games are independent.
Sam plays 20 games.

- (a) Find the probability that Sam wins 4 or more prizes. (2)

Tessa plays a series of games.

- (b) Find the probability that Tessa wins her 4th prize on her 20th game. (2)

Rama invites Sam and Tessa to play some new games of *Pento*.

They must pay Rama £1 for each game they play but Rama will pay them £2 for the first time they win a prize, £4 for the second time and £(2w) when they win their wth prize ($w > 2$)

Sam decides to play n games of *Pento* with Rama.

- (c) Show that Sam's expected profit is $\pounds \frac{1}{25}(n^2 - 16n)$ (6)

Given that Sam chose $n = 15$

- (d) find the probability that Sam does not make a loss. (4)

Tessa agrees to play *Pento* with Rama. She will play games until she wins r prizes and then she will stop.

- (e) Find, in terms of r , Tessa's expected profit. (4)

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Answer ALL questions. Write your answers in the spaces provided.

1. Bacteria are randomly distributed in a river at a rate of 5 per litre of water. A new factory opens and a scientist claims it is polluting the river with bacteria. He takes a sample of 0.5 litres of water from the river near the factory and finds that it contains 7 bacteria. Stating your hypotheses clearly test, at the 5% level of significance, whether there is evidence that the level of pollution has increased.

(5)

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4. A random sample of 100 observations is taken from a Poisson distribution with mean 2.3

Estimate the probability that the mean of the sample is greater than 2.5

(4)

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